

Computer Science & IT

Database Management System



Relational Model & Normal Forms

Lecture No. 02



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Recap of Previous Lecture



✓ Topic

Syllabus

✓ Topic

Introduction to DBMS

✓ Topic

Relational Database Model

Topics to be Covered



Topic

Relational Database Model



Topic

Functional dependency (FD)



Topic

Types of functional dependency





Topic : Relational database

Concept of Relational model is introduced by Dr Codd in 1970



- In relational model database is organized in the form of table { Collection of Rows a Columns }
- Each table of relational database is called a relation
- Dr. Codd defined 13 rules (numbered by 0 to 12) for a table to be defined as relation.



Topic : Relational database

eg:

Student

Attributes/fields

Records/tuples

Sid	Sname	Branch
S ₁	A	CS
S ₂	A	CS
S ₃	B	IT
S ₄	C	CS

- Degree / Arity :- No. of attributes in a relation is defined as degree / Arity of that relation
- Record / Tuple :- Each row of relation is called record / tuple
- Cardinality :- Number of tuples / records in a relation is defined as Cardinality of that Relation
- Relational Schema :- Relational Schema provides the abstract details of the relation
eg. Name-of-Relation (Name of 1st Attribute, Name of 2nd attribute,)
- Relational Instance :- If records are present in the relation, then that set of records is called relational instance of that time.



Topic : Relational database

→ Degree = 3 (No. of Columns)

→ Cardinality = 4 (No. of rows)

→ Relational Schema:

Student (Sid, Sname, Branch)

$\neq$ Student (Sname, Sid, Branch)

eg:

Student

Sid	Sname	Branch
S ₁	A	CS
S ₂	A	CS
S ₃	B	IT
S ₄	C	CS

Attributes/fields

$(a, b) \neq (b, a)$

Relational instance

$= \{ (S_1, A, CS), (S_2, A, CS), (S_3, B, IT), (S_4, C, CS) \}$



Topic : Functional dependency (FD)

Functional dependencies are used to specify the relationship among the attributes of a given relation.

Functional dependencies are denoted using the format

$$X \longrightarrow Y$$

Where X & Y are some set of attributes from the given relation

If $X \longrightarrow Y$ exist in relation R

then "X" is called determinant attribute set
& "Y" is called dependent attribute set



Topic : Functional dependency (FD)

- ☐ Functional dependency defines the relationship between two sets of attributes in a relational table.
- ☐ It states that the values of dependent attribute set can be completely determined by the values of determinant attributes set. In other words, if a determinant attribute's value is known, the value of the dependent attribute can be deduced.



Topic : Functional dependency (FD)

- Let R be the relational schema with X and Y as the attribute sets over relation R .

Functional dependency $X \rightarrow Y$ exists in R only if
For all pair of tuples $t_1, t_2 \in R$
If $t_1.X = t_2.X$ then $t_1.Y = t_2.Y$

IF, functional dependency $X \rightarrow Y$ exists in R
then (for all pair of tuples $t_1, t_2 \in R$
if $t_1.X = t_2.X$ then $t_1.Y = t_2.Y$)

P only if $Q \equiv$ if P then Q

P	Q	$(P \rightarrow Q)$
True	True	True

If $P \rightarrow Q$ is true
then is it necessary for $Q \rightarrow P$ to be true

Ans is NO.

∴ If, for all pair of tuples $t_1, t_2 \in R$
(if $t_1.X = t_2.X$ then $t_1.Y = t_2.Y$)

then functional dependency $X \rightarrow Y$ exists in R

Need not be true



Topic : Functional dependency (FD)

- If functional dependency $X \rightarrow Y$ exists in the relation
then,

Whenever values w.r.t. attribute set X are
same in any pair of tuples, then in those
tuples values of Y will also be same

it is the necessary condⁿ
for $X \rightarrow Y$ to exist in a
relation, but not sufficient

But if values of " Y " are same in
two tuples, then values of X need not
be same in those tuples

→ Consider the following relation: -

Sid	Sname	Branch
1	Ram	CS
2	Ram	CS
3	John	IT
4	Mohan	CS
5	Sai	IT

Student
Relation

If we know value of "Sid" then we can uniquely identify the "Sname" w.r.t. that "Sid"

∴ Functional dependency

$Sid \rightarrow Sname$

may exist in the relation

Necessary Condⁿ
holds true in this
relation w.r.t. FD

$Sid \rightarrow Sname$
x y

∴ $Sid \rightarrow Sname$
may hold in the
relation

→ Consider the following relation: -

Sid	Sname	Branch
1	Ram	CS
2	Ram	CS
3	John	IT
4	Mohan	CS
5	Sai	IT
Student Relation		
6	Ram	IT

In this specific relational instance, If we know the "Sname" then we can uniquely identify "Branch" w.r.t given Sname

∴ Sname → Branch may hold true

Necessary Condⁿ holds true in this relation w.r.t. FD

Sname → Branch

∴ Sname → Branch may hold in the relation

→ Consider the following relation: -

Sid	Sname	Branch
1	Ram	CS
2	Ram	CS
3	John	IT
4	Mohan	CS
5	Sai	IT

Student
Relation

In this specific relational instance, If we know the "Branch" of the student then we may not uniquely identify the "Sname" w.r.t. given "Branch".

∴ $\text{Branch} \rightarrow \text{Sname}$ does not hold (exists) in the given relation

Necessary Condⁿ does not hold true in this relation w.r.t. FD

$\text{Branch} \rightarrow \text{Sname}$

∴ FD, $\text{Branch} \rightarrow \text{Sname}$ can never exist in this relation



Topic : Functional dependency (FD)

- ❑ If necessary condition for functional dependency " $X \rightarrow Y$ " does not hold true based on given relation instance, then functional dependency " $X \rightarrow Y$ " can never exist in the given relation.
- ❑ Even if necessary condition for functional dependency " $X \rightarrow Y$ " does hold true based on given relation instance, then also we can not be sure whether functional dependency $X \rightarrow Y$ exists in the relation or not, because it is just the relational instance. ✓

#Q. Consider the following relational instance

	A	B	C
t_1	1	2	3
t_2	1	2	4
	2	2	1
	3	1	2
	4	1	2

Handwritten annotations on the table: Blue circles around (1,2,3) and (1,2,4). Yellow circles around (2,2,1), (3,1,2), and (4,1,2). A green arrow points from the first row to $A \not\rightarrow C$. A yellow arrow points from the last row to $BC \not\rightarrow A$.

$$A \not\rightarrow C$$

$$A \rightarrow B \quad \checkmark$$

$$AB \not\rightarrow C$$

$$BC \not\rightarrow A$$

Which of the following functional dependency **may hold true (not necessarily)** based on given relational instance.

a) $A \rightarrow C$

☒ b) $A \rightarrow B$

c) $AB \rightarrow C$

d) $BC \rightarrow A$

#Q.

Consider the following relational instance

A	B	C
1	1	1
1	2	2
2	4	3
3	3	4
4	1	5
5	3	6

No two values of 'C' are same
or we don't need to worry about other

Which of the following functional dependency **may hold true (not necessarily)** based on given relational instance.

~~a) $B \rightarrow C$~~

~~b) $A \rightarrow B$~~

~~c) $C \rightarrow B$~~

~~d) $B \rightarrow A$~~

~~e) $C \rightarrow A$~~

#Q. From the following instance of a relation schema $R(A,B,C)$, we can conclude that:

A	B	C
1	1	1
1	1	0
2	3	2
2	3	2

Necessary
Condⁿ is true
w.r.t.

$A \rightarrow B$

i.e. $A \rightarrow B$ may
hold, but we

Can not conclude that
 $A \rightarrow B$ holds in the relation

we can conclude, that

$B \rightarrow C$ can never hold in this relation

(A) A functionally determines B, and B functionally determines C

(B) A functionally determines B, and B does not functionally determine C

(C) B does not functionally determine C

(D) A does not functionally determine B, and B does not functionally determine C

Note: Functional dependencies that holds true in a relation are identified by database designer, and mostly they are given in the question



Topic : Types of Functional Dependency

* There are two types of functional dependencies

- ① Trivial functional dependency
- ② Non-trivial functional dependency

We can define a third type of functional dependency
Which is actually a mixture of trivial & non-trivial.
∴ We can call it "Semi-non-trivial functional dependency"



Topic : Trivial Functional Dependency

If X and Y are two sets of attributes over relation R such that $X \supseteq Y$,
then $X \rightarrow Y$ is called a trivial FD.

Every trivial FD possible
from the attributes of a relation
will always hold true in that relation
but they will not provide
any useful information

eg: $Sid, Sname \rightarrow Sid$
 $Sid, Sname \rightarrow Sname$
 $Sid, Sname \rightarrow Sid, Sname$



Topic : Non-trivial Functional Dependency

FD $X \rightarrow Y$ is called a non-trivial FD in relation R

Only if, (i) X and Y are non-empty set of attributes over relation R

and (ii) $X \cap Y = \emptyset$

eg. $R(\text{Sid}, \text{Sname}, \text{Branch})$ $\Rightarrow$ Possible non-trivial FDs w.r.t. attributes of rel R are

$\text{Sid} \rightarrow \text{Sname}$	$\text{Sname} \rightarrow \text{Branch}$	$\text{Sid} \rightarrow \text{Sname}, \text{Branch}$	$\text{Sid}, \text{Sname} \rightarrow \text{Branch}$
$\text{Sid} \rightarrow \text{Branch}$	$\text{Branch} \rightarrow \text{Sid}$	$\text{Sname} \rightarrow \text{Sid}, \text{Branch}$	$\text{Sid}, \text{Branch} \rightarrow \text{Sname}$
$\text{Sname} \rightarrow \text{Sid}$	$\text{Branch} \rightarrow \text{Sname}$	$\text{Branch} \rightarrow \text{Sid}, \text{Sname}$	$\text{Sname}, \text{Branch} \rightarrow \text{Sid}$

Every non-trivial FD need not hold true in the given relation



Topic : Semi non-trivial Functional Dependency

FD $X \rightarrow Y$ can be called a semi non-trivial FD

if and only if

(i) $X \not\equiv Y$

and (ii) $X \cap Y \neq \emptyset$

eg. $R(\text{Sid}, \text{Sname}, \text{Branch})$

$\text{Sid}, \text{Sname} \rightarrow \text{Sname}, \text{Branch}$

$\{\text{Sid}, \text{Sname}\} \not\equiv \{\text{Sname}, \text{Branch}\}$

$\{\text{Sid}, \text{Sname}\} \cap \{\text{Sname}, \text{Branch}\} = \{\text{Sname}\} \neq \emptyset$

Both condⁿ are satisfied,
∴ we can call it semi non-trivial FD.

14/04/20
#Q.



Consider the following relational instance

A	B	C
1	1	1
1	1	2
2	1	2
2	2	3
3	3	4

Find all non-trivial FDs which may hold true in the above relation based on given relational instance.

Q: Let $R(A, B, C, D)$ is a relation,
How many different non-trivial FDs can be
defined w.r.t. the attributes of relation R



2 mins Summary



Topic

Relational Database Model ✓

Topic

Functional dependency ✓

Topic

Types of functional dependency ✓

THANK - YOU